available to us and MAHs. This needs to change through collection of better data.

Medicine use in pregnancy remains a virtually uninformed decision with unknown long-term consequences. We must not forget, however, that the medical condition itself may be very harmful to the unborn child. Postauthorization information collection to help reduce uncertainty is challenging, given that registries may suffer from slow accrual, selection bias, loss to follow-up, and lack of follow-up beyond birth, whereas electronic health records may lack detail and statistical power, and, except for major teratogens, spontaneous reporting systems are notoriously insufficient in this area. Collaborative work on registries is ongoing.

As part of the EU Good Pharmacovigilance Practice guidelines,[4] further guidance is being developed. The aim is to provide a clearer overview and a more consistent, structured approach, for example on how to carefully balance the risks and benefits of treatment choices depending on the likelihood of pregnancy, as well as the need for and the methods of data collection in the postauthorization phase.

In addition, the EMA is preparing a pregnancy strategy paper focusing on collaboration with academia, patients, and the pharmaceutical industry to address the issues identified, to help enhance the protection of women and unborn children against harms arising from medicine use.

CONCLUSION

The EU regulatory framework on medicine use in pregnancy can be made a better fit for the purpose by using existing tools more effectively. This requires greater awareness and a concerted effort between regulators, academia, pharmaceutical industry, healthcare professionals, and the women themselves.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

DISCLAIMER

The views expressed in this article are the personal views of the authors and may not be understood or quoted as being made on behalf of or reflecting the position of the EMA, or one of the committees or working parties.

1. EUROmediCAT report. <http://euromedicat.eu/content/Final-Report-Executive-Summary-April-2015.pdf>. Accessed 4 Feb 2016.
2. Mitchell, A.A. Studies of drug-induced birth defects. In *Pharmacoepidemiology*, 5th ed. (Strom, B.L., Kimmel, S.E. & Hennessy, S., eds) (Hoboken, NJ: John Wiley & Sons, 2012).
3. Matsui, D. Ethics of studies of drugs in pregnancy. *Paediatr. Drugs.* **17**, 31–35 (2015).
4. European Medicines Agency Website. <www.ema.europa.eu>. Accessed 12 Feb 2016.
5. Daston, G.P. *et al*. Exposure-based validation list for developmental toxicity screening assays. *Birth Defects Res. B Dev. Reprod. Toxicol.* **101**:423–428 (2014).

Medication Use in Pregnancy and the Pregnancy and Lactation Labeling Rule

L Sahin[1], SC Nallani[2] and MS Tassinari[1]

On 30 June 2015, the US Food and Drug Administration Pregnancy and Lactation Labeling Rule (PLLR) took effect. This rule sets new and improved standards for the inclusion of information about the use of prescription drugs and biological products during pregnancy and lactation. The new labeling requirements have important implications for clinical pharmacology as there is a subheading that is dedicated to inclusion of clinical pharmacology information that inform dosing during pregnancy and the postpartum period, if available.

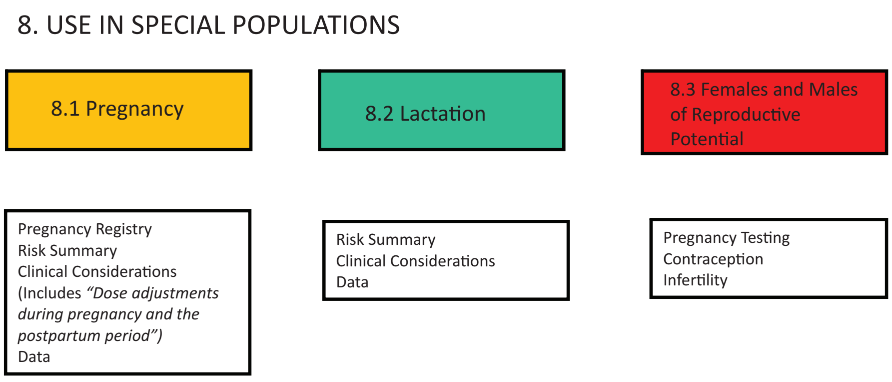
As a public health agency, part of the US Food and Drug Administration's mission is to ensure that the safety and effectiveness of drugs and biological products are adequately communicated in prescription drug and biological product labeling. Physiologic changes that occur during pregnancy may impact the pharmacokinetics (PK) and pharmacodynamics of drugs and biologics, and information on the correct dosing needed to achieve the desired therapeutic effect is important for prescribers. Providing evidence-based prescribing to pregnant women is an unmet medical need. Although medicines are routinely prescribed during pregnancy, they are commonly prescribed without important clinical data about dosing, PK, and safety in pregnant women. As part of agency efforts to improve the content and format of labeling, the US Food and Drug Administration issued a rule, referred to as the Pregnancy and Lactation Labeling Rule (PLLR), that specifically addresses labeling for pregnant and breastfeeding women.[1] This new rule provides an opportunity for inclusion of important clinical pharmacology information that may inform safe and effective prescribing in pregnant and lactating women.

The Pregnancy and Lactation Labeling Rule

The PLLR requires that all prescription drug and biological products, including

[1]Division of Pediatric and Maternal Health, Office of Drug Evaluation IV, Office of New Drugs, Center for Drug Evaluation and Research, US Food and Drug Administration, Silver Spring, Maryland, USA; [2]Division of Clinical Pharmacology 2, Office of Clinical Pharmacology, Office of Translational Sciences, Center for Drug Evaluation and Research, US Food and Drug Administration, Silver Spring, Maryland, USA. Correspondence: SC Nallani (Srikanth.Nallani@fda.hhs.gov)

doi:10.1002/cpt.380



A brief outline of the re-formatted labeling: In each subsection, clinical pharmacology data can be included.

See draft guidance: Pregnancy, Lactation, and Reproductive Potential: Labeling for Human Prescription Drug and Biological Products – Content and Format.[1] http://www.fda.gov/Drugs/DevelopmentApprovalProcess/DevelopmentResources/Labeling/ucm093307.htm

Figure 1 The pregnancy and lactation labeling final rule and changes to the prescription drug labeling.

generic labeling, remove the pregnancy letter categories A, B, C, D, and X from their product labeling. This letter category removal will be implemented over the next 2–4 years. These changes come after extensive reviews and discussions with key stakeholders. The pregnancy letter category system was overly simplistic because it did not capture the complexities of available risk information and risk-benefit considerations. The pregnancy letter categories were misinterpreted as a grading system, which resulted in prescribing decisions that were based on incorrect assumptions for the known risks of the drug. The PLLR requires drugs and biological products approved after 30 June 2001, to reformat the information found in the Pregnancy and Nursing Mothers subsections of product labeling. The old format and pregnancy letter category are replaced with narrative summaries of the risks of using a drug or biologic during pregnancy and lactation, a discussion of the data supporting that summary, and relevant information to help health care providers make prescribing decisions and counsel women about the use of drugs during pregnancy and lactation. The new labeling requirements describe, based on available information, a summary of the risks for the mother, the fetus, the breastfeeding child, and women and men of reproductive potential. Manufacturers are required to revise current labeling to reflect available data on use of the product during pregnancy and lactation and to update labeling when the information becomes outdated.

Description of the new labeling

Under current labeling, Section 8 – USE IN SPECIFIC POPULATIONS has specific subsections describing known information for pregnancy, lactation, and concerns for reproductive potential (**Figure 1**).[2] The Pregnancy (8.1) and Lactation (8.2) subsections include three headings: "Risk Summary," "Clinical Considerations," and "Data." These headings provide more detailed information regarding human, animal, and pharmacologic data on the use of the drug, and specific adverse reactions of concern for pregnant or breastfeeding women. The Pregnancy subsection also includes information for a pregnancy exposure registry for the drug when one is available. The Risk Summary is a narrative summary of the data designed to convey what is known about the potential risk of exposure for the drug during pregnancy. A new element for labeling under PLLR is the inclusion of statements when there is an absence of data to inform of the risk. The Clinical Considerations heading under Pregnancy includes information relevant to the use of a drug in pregnancy. Information that may be presented includes possible impacts of untreated disease so that prescribers and patients may make informed decisions. For example, a drug for the treatment of asthma could include a description of the effects of poorly controlled asthma in pregnancy that is associated with an increased risk of preeclampsia and small for gestational age. The Clinical Considerations section also includes the subheading, "Dose adjustments during pregnancy and the postpartum period," that is used to present any clinical pharmacology information that may inform dosing during pregnancy and the postpartum period. The Data section includes a concise description of the studies used to support the statements made in the Risk Summary. The Lactation subsection (8.2) is structured in a similar fashion and is intended to provide information about using the drug while breastfeeding, such as the amount of drugs in breast milk and potential effects on the breastfed infant. Data that inform this subsection of labeling may be derived, for example, from clinical lactation studies. The Clinical Considerations heading under Lactation includes information on minimizing exposure to the breastfed infant that relies on PK data from clinical studies.

The Females and Males of Reproductive Potential subsection (8.3) is a new subsection required under PLLR when relevant. This section presents information about pregnancy testing and contraception recommendations for suspected or known teratogens, and information about infertility effects. Data under this subsection may also include clinically relevant PK data on drug-drug interactions that may potentially affect contraception efficacy.

The new labeling requirements have implications for clinical pharmacology. It is important to understand the influence of pregnancy on the PK, and, where appropriate, the pharmacodynamics of drugs, as extensive physiologic changes in pregnancy can affect the PK/pharmacodynamic of drugs.[3] PK data along with safety information can help improve product labeling with regard to the dose adjustment as with any other specific population as well as guide women who wish to breastfeed their infants. Hence, PK studies in pregnant and lactating women who require life-saving medications may help derive important information to support safe use of drugs.[3] As an example, PK data for various antiretroviral drugs have been added to labeling to recommend dosing in pregnant and postpartum women.

In order to conduct studies in a vulnerable population, such as pregnant women, it is important to thoroughly plan the investigations with *in silico* methods. A well-known application of a population PK approach to amoxicillin PK changes that occur during pregnancy was published within the last decade.[4] A rare but real bioterrorism threat to the general public, including pregnant women and pediatric patients, due to anthrax was identified in the United States in the early 2000s. Based on the knowledge of amoxicillin PK in pregnancy and the simulations used to support different dosing regimens, amoxicillin dosing recommendations have been developed for prevention of anthrax in pregnant women.[4,5]

Updating the labeling

The new format for pregnancy and lactation labeling provides a means to update and communicate information to healthcare providers and their pregnant or lactating patients, but the data needed to make prescribing decisions are, for the most part, lacking. Addressing these gaps in data will require focused efforts on the part of multiple stakeholders, including the academic community, patient and advocacy groups, the pharmaceutical industry, and global regulatory authorities. For products in development, appropriate analyses should be anticipated and integrated into the drug development process. These analyses include studies needed to ensure understanding of potential drug-drug interactions between the drug and hormonal contraceptives, especially for drugs that present potential teratogenic risk. For approved products subject to the PLLR, data are reviewed and updated as part of the labeling reformatting.

To be effective and useful, drug and biological product labeling should clearly present current information. Currently marketed drugs subject to PLLR should have labeling updated to include data from available published epidemiologic, PK, and other clinical studies. The PLLR is an important step in increasing the availability of clear and current information for prescribers, pregnant women, or women who wish to breastfeed their infants.

DISCLAIMER

This article reflects the views of the authors and should not be construed to represent the US Food and Drug Administration's views or policies.

1. Content and Format of Labeling for Human Prescription Drug and Biological Products, Requirements for Pregnancy and Lactation Labeling (79 FR 72063, December 4, 2014). <https://www.gpo.gov/fdsys/granule/FR-2014-12-04/2014-28241> (2014).
2. Labeling requirements for new and more recently approved prescription drug products. 21 CFR 201.56 (d)(1) and 21 CFR 201.57(b)(9). <https://www.gpo.gov/fdsys/pkg/CFR-2015-title21-vol4/pdf/CFR-2015-title21-vol4-chapI.pdf> (2015).
3. Draft Guidance for Industry "Guidance for Industry Pharmacokinetics in Pregnancy—Study Design, Data Analysis, and Impact on Dosing and Labeling." <http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM072133.pdf> (2004).
4. Andrew, M.A. *et al*. Amoxicillin pharmacokinetics in pregnant women: modeling and simulations of dosage strategies. *Clin. Pharmacol. Ther.* **81**, 547–556 (2007).
5. Commentary on Non-Labeled Dosing of Oral Amoxicillin in Adults and Pediatrics for Post-Exposure Inhalational Anthrax. <http://www.fda.gov/drugs/emergencypreparedness/bioterrorismanddrugpreparedness/ucm072106.htm> (2015).